SPEAK ENERGY

EEBus Initiative e.V.
Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

EEBus High-Level Test Specification

Monitoring of Power Consumption

Version 1.0.0

Cologne, 2024-04-30

The EEBus concept was developed as part of E-Energy.

Supported by: Federal Ministry of Economics and Technology
on the basis of a decision by the German Bundestag

Table of contents

List of figures

List of tables

1 Introduction

1.1 Motivation

The objective of this document is to provide a basis for interoperability and comparable quality assurance for EEBus member companies and organizations regarding the various components used to meet the high-level requirements described in [MPC1.0.0].

Due to independent implementations of different vendors which lead to a high degree of complexity, the avoidance of implementation and interpretation errors is the goal for an interoperable system. Therefore, the technical specification for Monitoring of Power Consumption defines a generic framework for interoperable technology-independent implementations. To verify the desired interoperable behaviour between various applications, hardware and software components, the test specification is necessary.

This test specification defines a test suite of abstract test cases being relevant in order to ensure defined behaviour and to derive an agreed and common set of specific conformance tests.

The conformance tests described in this document are generic to apply to different implementations and therefore increase their ability of interoperability. Implementers should gain confidence that each implementation conforms the test specification.

1.2 Structure

Following objectives are identified in the scope of the test specification document:

- Specification of abstract test cases in agreement with the respective requirements
- Defined set of abstract test cases
- Standardized procedures within the abstract test cases
- Forming basic internals (due to black box testing) of each test in a defined test description

1.3 Limitations

Considering that it is impossible to define an exhaustive test suite due to independent implementations within different hardware approaches, these tests do not include the assessment of performance nor robustness or reliability of an implementation. The test specification itself cannot guarantee conformance to the technical specification since it detects errors rather than their absence. Hence, no guarantee of interoperability is given if the implementation is only conforming the test specification. It also focuses on the correct behavior rather than the verification of messages generated by a device under test (DUT). Due to this limitation, only limited tests might be defined in this test specification to verify that a DUT message complies with limits defined in a manufacturer's data sheet.

Furthermore, no requirements are specified that describe how a manufacturer enables the tester to change test case-specific values.

1.4 Contact information

Questions regarding the conformance process that are not addressed by this document should be directed to: info@eebus.org.

2 Scope

The purpose of the EEBus test specification for the technical specification of the Use Case "Monitoring of Power Consumption" (short name: MPC) is to verify the conformance to section 2 of [MPC1.0.0].

This document focuses on defining a conformance test suite which is then used as a necessary requirement for interoperability tests when dealing with projects using the EEBus technical specification for the Use Case "Monitoring of Power Consumption".

2.1 Overview

This document contains scope, referenced documents, abstract test cases (such as conformance or scenario) and test procedures for a Device Under Test (DUT) implementing the MPC according to the following document which define the Monitoring of Power Consumption and its underlying resources as defined in:

- EEBus_UC_TS_MonitoringOfPowerConsumption_V1.0.0 ([MPC1.0.0])

It provides a detailed description of each test, including as applicable, identification number(s) (ID) and used configurations.

In chapter 3, referenced documents are provided.

A detailed terms and abbreviations description can be found in chapter 4.

Chapter 5 defines the requirements which serve as the foundation on which the abstract test cases are based on.

In chapter 6, a framework for the abstract test cases and their execution is described. The sections contain general information and descriptions as well as the data sets to be used.

The chapters 7 and 8 contain the high-level abstract test cases for both Monitored Unit and Monitoring Appliance which describe the step-by-step actions, expected results and any special conditions necessary for testing.
Due to different approaches while using [MPC1.0.0] the abstract test cases are defined to be applicable for any specific implementation.

3 References

The following documents include essential guidelines and requirements which, through reference in this text, constitute guidelines and requirements of this document. In case of dated references, subsequent amendments to, or revisions of, any of these publications do not apply. Latest issue of undated references shall be used unless otherwise agreed.

3.1 EEBUS documents

[MPC1.0.0]	EEBus_UC_TS_MonitoringOfPowerConsumption_V1.0.0.pdf
[ParameterSheet]	EEBus_MPC_ParameterSheet_V1.0.0.xlsx

3.2 Normative references

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels (please see section 4.1 for details)

4 Terms, definitions and abbreviations

4.1 Terms and definitions

Within the scope of this document terms and definitions given in [MPC1.0.0] and the following apply.

As a source of common terminology for use in standardization please see the following databases of ISO, IEC and ISTQB:

- ISO Online Browsing Platform (OBP): https://www.iso.org/obp/ui
- IEC Electropedia: https://www.electropedia.org/
- ISTQB Glossary: https://glossary.istqb.org/en/search/

NOTE 1 Hyperlinks included in this document are subject to its validity at the time of publication, hence no guarantee can be given by the EEBus Initiative e.V. for its long-term validity.

4.1.1 Abstract Test Case

A test case without specific (implementation level) values for input data and expected results.

4.1.2 Black Box Testing

Black Box Testing is a software testing method in which the behaviour of a DUT is tested without having knowledge of internal code structure, implementation details and internal paths. This method mainly focuses on input and output of software applications.

4.1.3 Device Under Test (DUT)

A Device Under Test is a single component, an assembly or an appliance that is undergoing testing.

4.1.4 Expected result

The observable presumed behaviour of the DUT based on its test step.

4.1.5 MAY

Verbal form (as defined in [RFC2119]) used to indicate a course of action permissible and optional within the limits of the Use Case ([MPC1.0.0]).

4.1.6 Negative testing

Negative testing is a testing method which checks whether the application reacts as expected to an input or an operation that does not meet application requirements, e.g. by correctly rejecting invalid or improper data sets as an input.

NOTE 2 During test execution, values may be used (e.g. false data) that may violate the requirements defined in [MPC1.0.0].

4.1.7 Positive testing

Determines that the DUT works as expected by providing valid data sets as an input.

NOTE 3 This type of test behaviour is primarily defined in this document.

NOTE 4 During test execution, values may be used (e.g. false data) that may violate the requirements defined in [MPC1.0.0].

4.1.8 Pre-condition

The test steps needed to define a stable state of a test item and its environment prior to the test case execution.

4.1.9 SHALL

Verbal form (as defined in [RFC2119]) used to indicate requirements strictly to be followed in order to conform to the standard.

4.1.10 SHOULD

Verbal form (as defined in [RFC2119]) used to indicate that among several possibilities one is recommended as particularly suitable, or that a certain course of action is preferred but not necessarily required.

4.1.11 Specific Test Case

Within this document specific test cases are derived from abstract test cases by adding defined data sets.

4.1.12 Test suite

A set of test cases to be executed in a specific test run.

4.1.13 Verdict

Test verdicts are used to indicate the consequence/outcome of the execution of a test case. Possible outcome statements are "passed", "failed" or "not applicable" as defined in section 6.10.

4.2 Abbreviations

For the present document following abbreviations apply.

Term/ Abbreviation	Description
AC	Alternating Current
ATC	Abstract Test Case
DC	Direct Current
DUT	Device Under Test
MA	Monitoring Appliance
MPC	Monitoring of Power Consumption
MU	Monitored Unit
NT	Negative Testing
PT	Positive Testing
Ref No	Reference Number
STC	Specific Test Case
TC	Test Case
TestSpec	Test Specification
UC	Use Case

Table 1: Abbreviations

5 Requirements

5.1 Mapping of requirements

Within this document, unique identifiers are used primarily for each mandatory requirement extracted from [MPC1.0.0] allowing full traceability and easier management. Related recommended or optional requirements are provided with a sub-requirement ID following design guidelines of [MPC1.0.0]. In addition, markers used in [MPC1.0.0] are provided as well as the corresponding sections to ensure a quick search in the Use Case.

The short form of the identifier is as follows:

[MPC-TS-xxx/y]

Definition:

- "MPC" represents the use case abbreviation of the technical specification [MPC1.0.0];
- "TS" stands for Test Specification;
- "xxx" represents the unique number for the individual requirement; and
- "y" symbolizes a unique number and is only used for sub-requirements.

The notation of the Use Case references as well as an example is given below.

Ref No: marker, section(s)

Example:

Ref No: [MPC-021], 2.4.2 and 2.6.1

5.2 Requirements and definitions extracted from [MPC1.0.0]

[MPC-TS-001] Scenario 1: Monitor power
Total Active Power: The total active power of the MU as defined in [MPC1.0.0], Ref No: [MPC-011], 2.3 and 2.3.1.1.

[MPC-TS-002] Scenario 1: Monitor power
Phase-Specific Active Power: MUs that are connected to more than one phase may additionally provide the phase-specific values as defined in [MPC1.0.0], Ref No: [MPC-012], 2.3.1.1.

[MPC-TS-002/1] Phase-specific active power on phase A as defined in [MPC1.0.0], Ref No: [MPC-012/1], 2.3.1.1.

[MPC-TS-002/2] Phase-specific active power on phase B as defined in [MPC1.0.0], Ref No: [MPC-012/2], 2.3.1.1.

[MPC-TS-002/3] Phase-specific active power on phase C as defined in [MPC1.0.0], Ref No: [MPC-012/3], 2.3.1.1.

[MPC-TS-002/4] The phase-specific active power values MAY only be provided if the Actor MU is aware of its connected phases (this may be only one phase or two phases, too) as defined in [MPC1.0.0], 2.3.1.1.

[MPC-TS-003] Scenario 2: Monitor energy
Total Consumed Energy: The total energy consumed by the MU since installation or last reset as defined in [MPC1.0.0], Ref No: [MPC-021], 2.3 and 2.3.2.1.

[MPC-TS-003/1] SHALL only be provided if the MU is capable of consuming energy as defined in [MPC1.0.0], 2.3.2.1.

[MPC-TS-004] Scenario 2: Monitor energy
Total Produced Energy: The total energy produced by the MU since installation or last reset as defined in [MPC1.0.0], Ref No: [MPC-022], 2.3 and 2.3.2.1.

[MPC-TS-004/1] SHALL only be provided if the MU is capable of producing energy as defined in [MPC1.0.0], 2.3.2.1.

[MPC-TS-005] Scenario 3: Monitor current as defined in [MPC1.0.0], 2.3.3.

[MPC-TS-005/1] Phase-specific active AC current on phase A as defined in [MPC1.0.0], Ref No: [MPC-031/1], 2.3.3.1.

[MPC-TS-005/2] Phase-specific active AC current on phase B as defined in [MPC1.0.0], Ref No: [MPC-031/2], 2.3.3.1.

[MPC-TS-005/3] Phase-specific active AC current on phase C as defined in [MPC1.0.0], Ref No: [MPC-031/3], 2.3.3.1.

[MPC-TS-005/4] If the Actor Monitored Unit is aware of its connected phases, the phase specific current consumption SHOULD be provided as defined in [MPC1.0.0], 2.3.

[MPC-TS-005/5] Only the active current is considered in this Use Case as defined in [MPC1.0.0], Ref No: [MPC-030/1], 2.3.3.1.

[MPC-TS-005/6] Only rms values are considered as defined in [MPC1.0.0], Ref No: [MPC-030/2], 2.3.3.1.

[MPC-TS-006] Scenario 4: Monitor voltage
Phase-Specific AC Voltage: The actual phase-specific AC voltage values of the Actor MU are provided by this Scenario as defined in [MPC1.0.0], Ref No: [MPC-041], 2.3 and 2.3.4.1.

[MPC-TS-006/1] Phase-Specific AC Voltage between phase A and neutral as defined in [MPC1.0.0], Ref No: [MPC-041/1], 2.3.4.1.

[MPC-TS-006/2] Phase-Specific AC Voltage between phase B and neutral as defined in [MPC1.0.0], Ref No: [MPC-041/2], 2.3.4.1.

[MPC-TS-006/3] Phase-Specific AC Voltage between phase C and neutral as defined in [MPC1.0.0], Ref No: [MPC-041/3], 2.3.4.1.

[MPC-TS-006/4] Phase-Specific AC Voltage between phase A and phase B as defined in [MPC1.0.0], Ref No: [MPC-041/4], 2.3.4.1.

[MPC-TS-006/5] Phase-Specific AC Voltage between phase B and phase C as defined in [MPC1.0.0], Ref No: [MPC-041/5], 2.3.4.1.

[MPC-TS-006/6] Phase-Specific AC Voltage between phase C and phase A as defined in [MPC1.0.0], Ref No: [MPC-041/6], 2.3.4.1.

[MPC-TS-006/7] Only values related to the connected phases SHALL be delivered as defined in [MPC1.0.0], 2.3.4.1.

[MPC-TS-007] Scenario 5: Monitor frequency
AC Frequency: The Frequency, the MU measures at its AC connection as defined in [MPC1.0.0], Ref No: [MPC-051], 2.3 and 2.3.5.1.

[MPC-TS-008] Each MU’s measurement value has a state that defines whether the value is correct or is to be ignored by the MA as defined in [MPC1.0.0], 2.5.2.

[MPC-TS-008/1] Out of range: value is out of range and SHALL be ignored by the MA as defined in [MPC1.0.0], 2.5.2.

[MPC-TS-008/2] Error: value is erroneous and SHALL be ignored by the MA as defined in [MPC1.0.0], 2.5.2.

[MPC-TS-009] Measured electrical current, active power and exchanged energy are expressed with positive values in case of energy consumption whereas negative values are used in case of energy production as defined in [MPC1.0.0], Ref No: [MPC-001], 2.5.1.

[MPC-TS-010] Voltages are measured independent of the energy direction as defined in [MPC1.0.0], Ref No: [MPC-002], 2.5.1.

[MPC-TS-011] MUs may be connected to more than one phase of the grid connection point of the house or premises. In this case, the power measurands can be provided for the individual phases, but a device is not obliged to offer these phase-specific values (the total-value is independent from that choice and SHALL always be provided) as defined in [MPC1.0.0], 2.1.

[MPC-TS-012] A CEM SHALL NOT act as Actor Monitored Unit within this Use Case as defined in [MPC1.0.0], 2.1.

282

5.3 Supplementary requirements

[MPC-TS-013] For appliances that request/send data via interval (polling), the corresponding ATCs are no longer considered "recommended" but "mandatory" and SHALL be executed.

[MPC-TS-014] For appliances that request/send data via notification, the corresponding ATCs are no longer considered "recommended" but "mandatory" and SHALL be executed.

6 Test suite conventions

6.1 General information

This chapter defines all conventions that are relevant for conformance tests of DUTs implementing [MPC1.0.0].

6.2 Conceptual test process

Figure 1 illustrates the conceptual test process to provide an overview and understanding of the test infrastructure elements required to perform tests regarding the Monitoring of Power Consumption Use Case. It represents a simplification of the structural relationships of the components and can be thought of conceptually as a set of interacting process steps. Each step corresponds to a particular aspect of functionality in the test system. The explanation of the conceptual test process elements, e.g. the [ParameterSheet] as a base document for the specific test cases, can be found in the following sections.

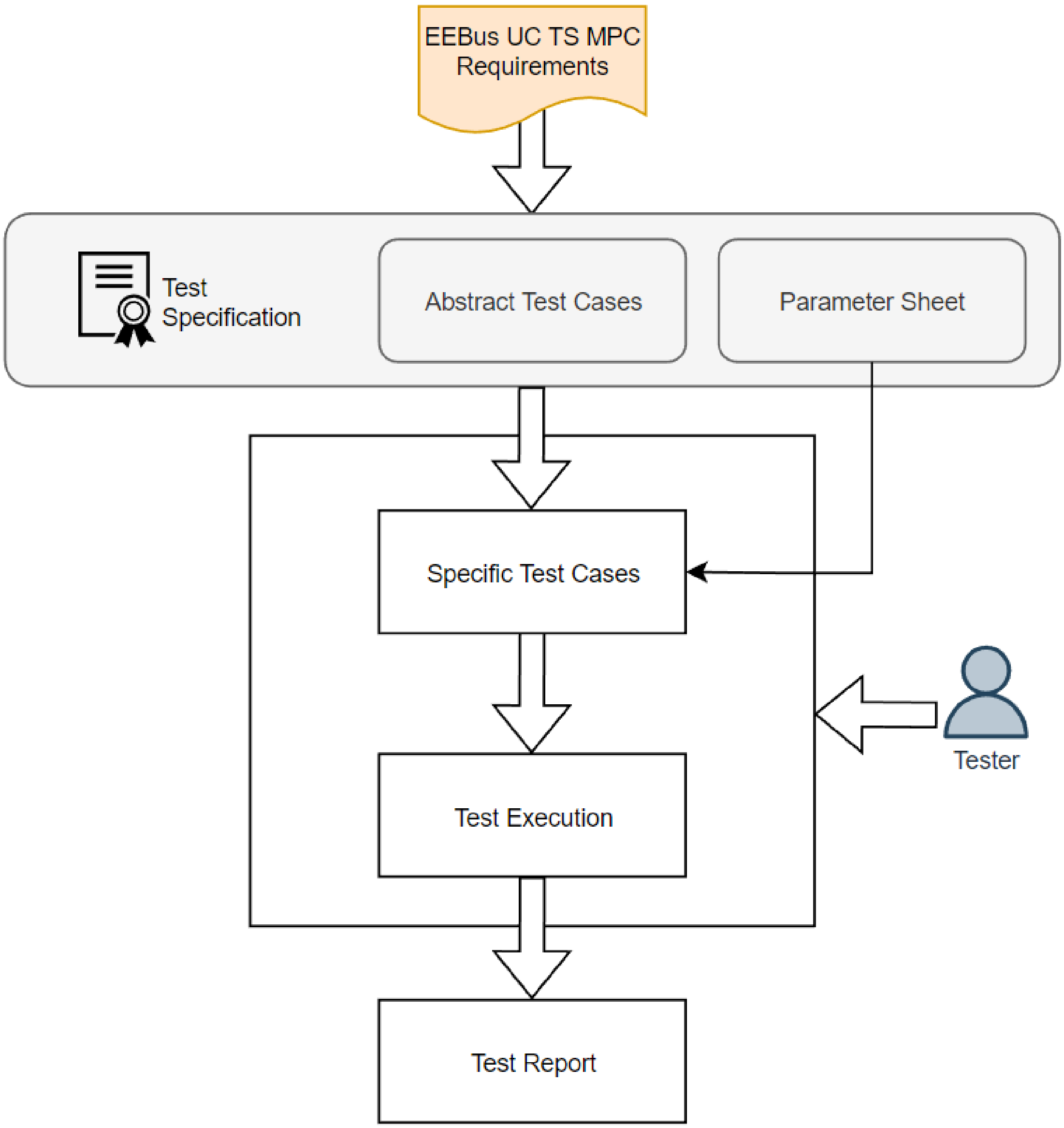

Figure 1: Conceptual test process

6.2.1 Test specification

This document defines ATCs derived from requirements extracted from [MPC1.0.0] (see section 5).

6.2.1.1 Parameter sheet – specific test cases

The [ParameterSheet] "Specific Test Cases" worksheet, provided in a separate document, is intended to support the test engineer in creating STCs derived from ATCs. Furthermore, it contains all device specific information of the manufacturer.

Cells that need to be filled in by the manufacturer/test engineer are highlighted with a white background whereas fields with a grey background should not be filled in manually. Each value entered in the corresponding cell above the table for the STCs, is completed in the STCs by automatic filling using a formula.

NOTE 5 Due to different values used in different test steps for the same parameter in one STC, some test cases are listed in two lines and marked with a suffixed STC ID, e.g. 1.1 and 1.2.

NOTE 6 For more information on using the [ParameterSheet], see the "Legend" worksheet.

6.2.2 Specific test cases

Due to the necessary variation of test input data (see section 6.11), ATCs need to be transferred to specific test cases. These data sets are manufacturer dependent as applications (thus the DUT) can be different in many details, e.g. devices may vary in their applicable power range. Specific test cases based on the ATCs are derived by using test input values of the [ParameterSheet]. Thus, test input values shall be defined in the [ParameterSheet] by the test engineer.

NOTE 7 For more information on using the [ParameterSheet], see the "Legend" worksheet.

6.2.3 Test execution

After the STCs were created, entry conditions for the test execution are met and the test execution phase begins. Tests shall be conducted as per the defined test cases. This step involves comparing actual results with expected results.

As described in section 1.3, it is not specified how a test engineer can verify both the limits and the states a DUT is in during test execution. Therefore, the manufacturer must ensure, e.g. via debug outputs or a graphical interface, that the required information is made available to the test engineer.

6.2.4 Test report

A test report is a document containing information about the performed tests and collected metrics like failed or passed test cases, test case coverage, detected bugs, spent time and results of the test runs. This report may contain manufacturer-specific information and is subject to the manufacturer.

6.3 Test suite identifiers

6.3.1 Abstract test case identifier

According to the naming convention defined by ETSI TS 102 869-3 V1.5.1 (European Telecommunications Standards Institute – https://www.etsi.org/), this convention is followed and applied to ATCs as shown in Table 2.

Element	Naming convention	Prefix	Example identifier
Abstract test case name	The identifier begins with ATC. It is mandatory to use underscores as separators within an identifier.	ATC	ATC_SCE1_PT_MAPhaseA_001

Table 2: Naming convention for abstract test case names

The identifier of the ATC is built according to Table 3 within this document.

ATC identifier: <prefix>_<dom>_<tot>_<ctx>_<xxx>

Identifier	Value	Description
<prefix>		see Table 2
<dom>		Domain
	SCE{n}	UC scenario, e.g. SCE1
	COM	Common
<tot>		Type of testing
	PT	Positive testing
	NT	Negative testing
<ctx>		Context
	{name}	e.g. MAPhaseA
<xxx>		Number
	{nnn}	Unique number from 001 to 999

Table 3: Naming convention for abstract test case identifiers

6.3.2 Test configuration identifier

The short form of the test configuration is as follows:

<prefix>_<actor>_<config>

Examples:

MA: CF_MA_ConnectionEstablished
MU: CF_MU_ConnectionEstablished

The identifier is built according to Table 4.

Identifier	Value	Description
<prefix>		Test configuration
	CF	Configuration
<actor>		Actor
	MA	Monitoring Appliance
	MU	Monitored Unit

<config>		Test configuration
	{name}	MA: ConnectionEstablished MU: ConnectionEstablished

Table 4: Naming convention for test configurations

Test configurations are defined in section 6.5.

6.4 Abstract test case description

The template for an abstract test case (ATC) is given below.

ATC ID	The ATC ID is a unique identifier for an ATC according to its definition in chapter 6.3.1.
Description	A brief description of the test objective is given here in accordance with the Use Case.
Referenced Requirement(s)	The referenced requirement(s) refers to requirements stated in section 5.2 of this document. The requirements are referenced according to the format defined in 5.1.
Pre-condition	The pre-condition provides a short description of the state the DUT is in before the actual ATC is executed. This may contain a test configuration which is referenced according to the format defined in section 6.5. Test step 1 is executed immediately after the pre-condition has been fulfilled.
Test variation	Due to variations within the data sets in section 6.11 the ATC defines the number of STCs. In addition, possible value states (see section 6.11.2) and the data to be used for STCs (see section 6.11.3) are listed here. Detailed information about deriving STCs can be found in section 6.9.
Execution	This element describes the test steps dealing with the actions to be performed and what is observed or measured during execution. The steps are performed within a certain time, unless otherwise described. There is no delay between the individual steps intended.
Expected result	Describes the observable expected results of the test steps during and after the Execution. For the execution of the STCs, it is mandatory to be able to clearly identify these states in detail. The overall verdict is only 'passed' if there is no failed result for an intermediate test or test step.

Table 5: Abstract test case description template

6.4.1 Abstract test case example

For clarification, individual components of an ATC are explained in more detail in the following example.

ATC ID	ATC_SCE1_PT_MAExample_001
Description	This test shall ensure that the MA receives the data.
Referenced	[MPC-TS-002]

Requirement(s)		
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce *[1]	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the momentary power consumption/production.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production.	The MA receives the data for the momentary power consumption/production within 120 seconds since the value changed.

Table 6: Abstract test case example

*[1]: The ATC shall be executed with both monitoring value states of the data point as well as both energy directions since it is marked with (all).

6.5 Test configurations

The test configuration reflects fixed conditions including "Monitoring of Power Consumption" Use Case states in which the MU can be as well as a defined MA behaviour before and during an ATC execution. In order to implement ATCs from a defined behaviour, these configurations are used as pre-conditions.

Within a configuration used in the pre-condition of an ATC, starting the corresponding timeouts must also be considered. The maximum possible time must be available for each configuration of the respective actor. It does not matter whether this is achieved via debug interface, exact logging or triggering a message by the tester.

6.5.1 Behaviour of the DUT

The configuration of a test object determines the start condition and the immediate start of test step 1. The configuration must not be delayed in such a way that the timeouts defined in the test steps are violated.

The rationale is that a configuration may be associated with a state "A" of an actor and a timer defined in [MPC1.0.0]. This means that it can be defined in [MPC1.0.0] that a certain actor (DUT) starts a timer as soon as it reaches the "A" state and that the actor either switches to the "B" state as soon as the timer ends or that it switches to another state "C" if, for example, it receives a command before the timer ends.

Typically, two test cases are defined in a test specification that cover such transitions: a test case that verifies the change to state "C" by a command before the timer ends; another test case that verifies the change to state "B" as soon as the timer ends. The execution of these test cases depends on the control of the timer. This means that the configuration of the DUT for such test cases corresponds to state "A" with a just initialised timer and that the execution of test step 1 starts immediately. In practice, there may be a certain delay between the configuration and the actual execution of test step 1. However, these delays must be short enough to allow a reliable verification of the time-controlled conditions and behaviour.

6.5.2 Behaviour of the tester

The configuration of the tester must support the test of the DUT. This includes taking the configuration of the DUT into account, particularly with regard to the DUT's timers. In general, the configuration of the tester is defined in such a way that it is fully functional in accordance with [MPC1.0.0]. However, the execution of the tests allows two test paradigms:

1. Full-featured actor as tester
2. Simplified actor as tester

In the first paradigm, the tester is or behaves like a fully functional implementation of the actor as specified in [MPC1.0.0]. The configuration of the tester must therefore be applied as specified in the test specification.

In the second paradigm, the tester focuses on the verification of the expected results described. In addition, it performs the necessary actions required to maintain the connection and the execution of the test steps.

NOTE 8 Test paradigm 1 might detect invalid behaviour of a DUT by detecting invalid messages that are not explicitly described in a test step. This behaviour should be logged as an error for later verification.

6.5.3 Monitoring Appliance

This document covers the different roles an MA will be used in when performing the described ATCs. The actor MA can either be used as the tester or the DUT. Depending on the corresponding test configuration a set of components and interactions for the MA are defined.

Valid test configurations for the MA within this document are shown in Table 7.

Configuration	Description for the MA *¹	Description for the MA as simplified tester *²
CF_MA_ConnectionEstablished	The MA is already connected to the MU. Necessary steps for establishing, as well as maintaining the connection during test execution are not explicitly described in the test steps. If an MA action is required, it is described within the test steps.	Same as in the description for the MA.

Table 7: MA test configurations

***[1]**: Applies for the MA as DUT or for a fully-featured MA as tester, see section 6.5.2.

***[2]**: Applies for the MA as simplified tester, see section 6.5.2.

6.5.4 Monitored Unit

These test configurations reflect fixed conditions, according to [MPC1.0.0], which the MU has in each state.

Valid test configurations for the MU as the tester and the DUT within this document are shown below.

Configuration	**State for the MU *[1]**	**Behaviour of the MU as tester *[2]**
CF_MU_ConnectionEstablished	The MU is already connected to the MA. Necessary steps for establishing, as well as maintaining the connection during test execution are not explicitly described in the test steps. If a MU action is required, it is described within the test steps.	Same as in the description for the MU.

Table 8: MU test configurations

***[1]**: Applies for the MU as DUT or for a fully-featured MU as tester, see section 6.5.2.

***[2]**: Applies for the MU as simplified tester, see section 6.5.2.

6.6 Timeouts and timings

For the scope of this document internal timings of any DUT cannot be tested explicitly in black box testing since they are not accessible by the tester. Therefore, only timeouts from the tester's perspective can be accurately measured and are considered in this document.

In order to gain a common understanding of how to handle timeouts during test execution, it is formally defined as follows:

- The corresponding timeout starts after the previous test step execution or the test step execution specified in the timeout identifier prefix.
- If the test step(s) end within the appropriate time and it corresponds to the expected result, the test continues. *[1]
- In case the test step(s) do not end by the time defined by the corresponding timeout value, the test verdict is set to 'false'.

*[1]: This formal definition assumes that the duration of the determination of a test result does not delay the execution of the next test step. In fact, for most test steps, it is acceptable to record relevant time-stamped data and determine the test result of each test step later during test execution or even after all test steps have been executed. Such "delayed" test step verification is NOT possible in any of the following cases:

- The next test step execution depends on a previous test result.
  Examples:
  - Depending on a test result, the next test step is either test step x or test step y.
  - The next test step contains a conditional statement that depends on a previous test result.
- Executing the next test step without considering the current test result would endanger any device.
  Both the manufacturer of a DUT and the test engineer are responsible for disclosing such constraints of their equipment prior to test execution. The test engineer is then responsible for taking these constraints into account during test execution.

Regardless of immediate or delayed test step verification, the first test step that deviates from the expected result is considered as last test step executed. I.e. even if some further test steps have been executed, they are considered not to have been executed.

6.7 Monitoring data update

[MPC1.0.0] does not define any conditions on when the MU must update its data points. Chapter 2 of [MPC1.0.0] also does not prescribe specific mechanisms for exchanging data updates. However, since data point monitoring is an essential part of [MPC1.0.0], this test specification also aims to verify the exchange of data updates. In practice, this is realized with the following two mechanisms:

- Polling: The MA requests data of the MU periodically by an implemented interval.
- Notification: New data is notified by the MU as soon as the data of a data point changes or is updated.

In order to be able to guarantee a defined test behavior, the following conditions for notifications apply to all test cases: the manufacturer must specify in the [ParameterSheet] when a value change of the respective data point is notified. This message must be transmitted within 120 seconds.

6.8 Test case execution

Within this document each test case executes a specific test behaviour and verifies the behaviour of the DUT according to its message-based response at any point in time. To ensure interoperability, each test case starts and ends in a defined and recoverable state of both the tester and the DUT. Therefore, the following steps are defined.

- *Pre-condition:* The tester and the DUT should be set to a configuration that allows communication between the two devices.
  If a pre-condition has been defined, both the tester and the DUT initialize to a known and stable state. This means that tester and DUT must be in a valid configuration before the test execution is performed.
- *Test behaviour during test execution:* In order to achieve the test objective, the test behaviour describes the necessary test steps during the test execution. At the end of each test step, the result is evaluated and a test verdict is assigned (see section 6.6 for further

details, especially regarding the results of the delayed test steps). The actions to be performed are explained below.

a. In case of test steps concerning notification, the relevant notify-timeout must be initialized on both tester and DUT sides.
b. MU as DUT:
    - Polling: Send a data request to the DUT.
    - Notification: Receive and verify the notification from the DUT after a value change.

   MA as DUT:
    - Polling: Receive and verify the data request of the DUT.
    - Notification: Send the notify to the DUT after a value change.
c. Stop timeout.
d. Assign a test verdict for the test behaviour.

6.9 Test case variation

Some ATCs (see chapters 7 and 8) specify one or more data sets for the "test variation" row that can or have to be used to derive STCs. For an example, see section 6.4.1.

The following keywords define how many of the listed data sets have to be used: *[1]

1. (all): If a data set is extended with "(all)" in the ATC, all possible values shall be used per STC. This extension is decisive for the number of STCs to be derived.
2. (any): If a data set is extended with "(any)" in the ATC, at least one of the possible values shall be used for all STCs.

*[1]: The information in the test case variation does not refer to heartbeats.

6.10 Test verdict

The test verdicts defined in this document are listed in Table 9.

Verdict statement	Definition
passed	This verdict statement shall be used when the DUT shows the correct behaviour with respect to the relevant requirements.
failed	This verdict statement shall be used when the DUT shows the wrong behaviour with respect to the relevant requirements.
not applicable	This verdict statement shall be used when the test case is not relevant to the DUT to be tested.

Table 9: Test verdicts

An overview of the test case coverage regarding its verdict can be found in section 6.13.

6.11 Data sets

The data sets defined in this section contain various values for the [MPC1.0.0] scenarios as well as possible combinations in which these values can be sent or received. In order to ensure the greatest possible test coverage, the ATCs define which values and combinations have to be used (see section 6.9).

6.11.1 Message handling

With reference to section 6.8, messages are either sent by the tester and received by the DUT or vice versa. Due to the defined roles within [MPC1.0.0], the following two message handling rules emerge depending on the type of DUT.

1. *MU as DUT:* The tester (MA) either sends a request to the DUT and expects the corresponding response message from the DUT or gets notified when a value of a data point has changed as described in section 6.8 step 2.
2. *MA as DUT:* Either the tester (MU) sends the corresponding data to the DUT after receiving a read request from it, or the tester sends data to the DUT after a value change via notification.

6.11.2 Monitoring value states

With respect to the data, test cases ensure that the received values are either correct, out of range or erroneous. Therefore, the manufacturer needs to specify the data for each data point in the [ParameterSheet], considering each state these values can take.

Based on black box testing, the assumption is that MonitoringState_01, i.e. the correct value, can be used by both the MU and the MA as DUT. MonitoringState_02 (out of range) and MonitoringState_03 (erroneous) are tested exclusively in ATCs with the MA as DUT.

To ensure conformance to the requirements defined in [MPC1.0.0], the following value states are used as a variation to derive STCs:

- MonitoringState_01: The state shall be normal (the value is correct).
- MonitoringState_02: The state shall be out of range.
- MonitoringState_03: The state shall be error.

Within the test variation of an ATC, the following statement can be found: "Monitoring value states (all):". A more detailed description of how to understand this statement can be found in section 6.9. For an example, see section 6.4.1.

6.11.3 Monitoring values

Listed below are the data points that have a positive value in case of energy consumption and a negative value in case of energy production. In both cases, the values are considered "correct" regardless of the state of the transmitted value as long as they comply with the sign rule of the UC.

- Total Active Power
- Phase-Specific Active Power
- Total Consumed Energy
- Total Produced Energy

- Phase-Specific AC Current

The data points, which are measured independently of the energy direction and are therefore always positive, are as follows:

- Phase-Specific AC Voltage
- AC Frequency

6.11.4 Energy directions

If the behaviour of a monitoring value is to be observed as a dependance of the energy direction of the actor Monitored Unit (MU), the required energy directions are listed as a variation for deriving STCs:

- consume
- produce

Within the test variation of an ATC, the following statement can be found: "Energy directions (all):". A more detailed description of how to understand this statement can be found in section 6.9. For an example, see section 6.4.1.

6.12 Requirement coverage

Table 10 provides an overview of the requirements covered in the test cases and the requirements that are e.g. out of scope according to the conventions defined in this document or cannot be tested in black box tests.

Requirement ID	Comment/ATC ID(s)
[MPC-TS-001]	ATC_SCE1_PT_MUTotalActivePower_001, ATC_SCE1_PT_MATotalActivePower_001
[MPC-TS-002]	ATC_SCE1_PT_MUPhaseActivePower_001, ATC_SCE1_PT_MUPhaseActivePower_002, ATC_SCE1_PT_MUPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_001, ATC_SCE1_PT_MAPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_005
[MPC-TS-002/1]	ATC_SCE1_PT_MUPhaseActivePower_001, ATC_SCE1_PT_MAPhaseActivePower_001
[MPC-TS-002/2]	ATC_SCE1_PT_MUPhaseActivePower_002, ATC_SCE1_PT_MAPhaseActivePower_003
[MPC-TS-002/3]	ATC_SCE1_PT_MUPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_005
[MPC-TS-002/4]	ATC_SCE1_PT_MUPhaseActivePower_001, ATC_SCE1_PT_MUPhaseActivePower_002, ATC_SCE1_PT_MUPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_001, ATC_SCE1_PT_MAPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_005,

[MPC-TS-003]	ATC_SCE2_PT_MUTotalConsumedEnergy_001, ATC_SCE2_PT_MUTotalConsumedEnergy_002, ATC_SCE2_PT_MATotalConsumedEnergy_001
[MPC-TS-003/1]	ATC_SCE2_PT_MUTotalConsumedEnergy_001, ATC_SCE2_PT_MUTotalConsumedEnergy_002, ATC_SCE2_PT_MATotalConsumedEnergy_001
[MPC-TS-004]	ATC_SCE2_PT_MUTotalProducedEnergy_001, ATC_SCE2_PT_MUTotalProducedEnergy_002, ATC_SCE2_PT_MATotalProducedEnergy_001
[MPC-TS-004/1]	ATC_SCE2_PT_MUTotalProducedEnergy_001, ATC_SCE2_PT_MUTotalProducedEnergy_002, ATC_SCE2_PT_MATotalProducedEnergy_001
[MPC-TS-005]	ATC_SCE3_PT_MUActiveACCurrent_001, ATC_SCE3_PT_MUActiveACCurrent_002, ATC_SCE3_PT_MUActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_001, ATC_SCE3_PT_MAActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_005
[MPC-TS-005/1]	ATC_SCE3_PT_MUActiveACCurrent_001, ATC_SCE3_PT_MAActiveACCurrent_001
[MPC-TS-005/2]	ATC_SCE3_PT_MUActiveACCurrent_002, ATC_SCE3_PT_MAActiveACCurrent_003
[MPC-TS-005/3]	ATC_SCE3_PT_MUActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_005
[MPC-TS-005/4]	ATC_SCE3_PT_MUActiveACCurrent_001, ATC_SCE3_PT_MUActiveACCurrent_002, ATC_SCE3_PT_MUActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_001, ATC_SCE3_PT_MAActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_005
[MPC-TS-005/5]	Tested indirectly with the information in the [ParameterSheet].
[MPC-TS-005/6]	Out of scope
[MPC-TS-006]	ATC_SCE4_PT_MUACVoltage_001, ATC_SCE4_PT_MUACVoltage_002, ATC_SCE4_PT_MUACVoltage_003, ATC_SCE4_PT_MUACVoltage_004, ATC_SCE4_PT_MUACVoltage_005, ATC_SCE4_PT_MUACVoltage_006, ATC_SCE4_PT_MAACVoltage_001, ATC_SCE4_PT_MAACVoltage_003, ATC_SCE4_PT_MAACVoltage_005, ATC_SCE4_PT_MAACVoltage_007, ATC_SCE4_PT_MAACVoltage_009, ATC_SCE4_PT_MAACVoltage_011
[MPC-TS-006/1]	ATC_SCE4_PT_MUACVoltage_001, ATC_SCE4_PT_MAACVoltage_001
[MPC-TS-006/2]	ATC_SCE4_PT_MUACVoltage_002, ATC_SCE4_PT_MAACVoltage_003
[MPC-TS-006/3]	ATC_SCE4_PT_MUACVoltage_003, ATC_SCE4_PT_MAACVoltage_005
[MPC-TS-006/4]	ATC_SCE4_PT_MUACVoltage_004, ATC_SCE4_PT_MAACVoltage_007

[MPC-TS-006/5]	ATC_SCE4_PT_MUACVoltage_005, ATC_SCE4_PT_MAACVoltage_009
[MPC-TS-006/6]	ATC_SCE4_PT_MUACVoltage_006, ATC_SCE4_PT_MAACVoltage_011
[MPC-TS-006/7]	ATC_SCE4_PT_MUACVoltage_001, ATC_SCE4_PT_MUACVoltage_002, ATC_SCE4_PT_MUACVoltage_003, ATC_SCE4_PT_MUACVoltage_004, ATC_SCE4_PT_MUACVoltage_005, ATC_SCE4_PT_MUACVoltage_006, ATC_SCE4_PT_MAACVoltage_001, ATC_SCE4_PT_MAACVoltage_003, ATC_SCE4_PT_MAACVoltage_005, ATC_SCE4_PT_MAACVoltage_007, ATC_SCE4_PT_MAACVoltage_009, ATC_SCE4_PT_MAACVoltage_011
[MPC-TS-007]	ATC_SCE5_PT_MUFrequency_001, ATC_SCE5_PT_MAFrequency_001
[MPC-TS-008]	Tested only for the MA as DUT. ATC_SCE1_NT_MATotalActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_004, ATC_SCE1_NT_MAPhaseActivePower_006, ATC_SCE2_NT_MATotalConsumedEnergy_002, ATC_SCE2_NT_MATotalProducedEnergy_002, ATC_SCE3_NT_MAActiveACCurrent_002, ATC_SCE3_NT_MAActiveACCurrent_004, ATC_SCE3_NT_MAActiveACCurrent_006, ATC_SCE4_NT_MAACVoltage_002, ATC_SCE4_NT_MAACVoltage_004, ATC_SCE4_NT_MAACVoltage_006, ATC_SCE4_NT_MAACVoltage_008, ATC_SCE4_NT_MAACVoltage_010, ATC_SCE4_NT_MAACVoltage_012, ATC_SCE5_NT_MAFrequency_002
[MPC-TS-008/1]	Tested only for the MA as DUT. ATC_SCE1_NT_MATotalActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_004, ATC_SCE1_NT_MAPhaseActivePower_006, ATC_SCE2_NT_MATotalConsumedEnergy_002, ATC_SCE2_NT_MATotalProducedEnergy_002, ATC_SCE3_NT_MAActiveACCurrent_002, ATC_SCE3_NT_MAActiveACCurrent_004, ATC_SCE3_NT_MAActiveACCurrent_006, ATC_SCE4_NT_MAACVoltage_002, ATC_SCE4_NT_MAACVoltage_004, ATC_SCE4_NT_MAACVoltage_006, ATC_SCE4_NT_MAACVoltage_008, ATC_SCE4_NT_MAACVoltage_010, ATC_SCE4_NT_MAACVoltage_012, ATC_SCE5_NT_MAFrequency_002

[MPC-TS-008/2]	Tested only for the MA as DUT. ATC_SCE1_NT_MATotalActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_002, ATC_SCE1_NT_MAPhaseActivePower_004, ATC_SCE1_NT_MAPhaseActivePower_006, ATC_SCE2_NT_MATotalConsumedEnergy_002, ATC_SCE2_NT_MATotalProducedEnergy_002, ATC_SCE3_NT_MAActiveACCurrent_002, ATC_SCE3_NT_MAActiveACCurrent_004, ATC_SCE3_NT_MAActiveACCurrent_006, ATC_SCE4_NT_MAACVoltage_002, ATC_SCE4_NT_MAACVoltage_004, ATC_SCE4_NT_MAACVoltage_006, ATC_SCE4_NT_MAACVoltage_008, ATC_SCE4_NT_MAACVoltage_010, ATC_SCE4_NT_MAACVoltage_012, ATC_SCE5_NT_MAFrequency_002
[MPC-TS-009]	ATC_SCE1_PT_MUTotalActivePower_001, ATC_SCE1_PT_MUPhaseActivePower_001, ATC_SCE1_PT_MUPhaseActivePower_002, ATC_SCE1_PT_MUPhaseActivePower_003, ATC_SCE2_PT_MUTotalConsumedEnergy_001, ATC_SCE2_PT_MUTotalConsumedEnergy_002, ATC_SCE2_PT_MUTotalProducedEnergy_001, ATC_SCE2_PT_MUTotalProducedEnergy_002, ATC_SCE3_PT_MUActiveACCurrent_001, ATC_SCE3_PT_MUActiveACCurrent_002, ATC_SCE3_PT_MUActiveACCurrent_003, ATC_SCE1_PT_MATotalActivePower_001, ATC_SCE1_PT_MAPhaseActivePower_001, ATC_SCE1_PT_MAPhaseActivePower_003, ATC_SCE1_PT_MAPhaseActivePower_005, ATC_SCE2_PT_MATotalConsumedEnergy_001, ATC_SCE2_PT_MATotalProducedEnergy_001, ATC_SCE3_PT_MAActiveACCurrent_001, ATC_SCE3_PT_MAActiveACCurrent_003, ATC_SCE3_PT_MAActiveACCurrent_005
[MPC-TS-010]	ATC_SCE4_PT_MUACVoltage_001, ATC_SCE4_PT_MUACVoltage_002, ATC_SCE4_PT_MUACVoltage_003, ATC_SCE4_PT_MUACVoltage_004, ATC_SCE4_PT_MUACVoltage_005, ATC_SCE4_PT_MUACVoltage_006, ATC_SCE4_PT_MAACVoltage_001, ATC_SCE4_PT_MAACVoltage_003, ATC_SCE4_PT_MAACVoltage_005, ATC_SCE4_PT_MAACVoltage_007, ATC_SCE4_PT_MAACVoltage_009, ATC_SCE4_PT_MAACVoltage_011
[MPC-TS-011]	Out of scope
[MPC-TS-012]	Out of scope

[MPC-TS-013]	ATC_COM_PT_MUPolling_001, ATC_SCE2_PT_MUTotalConsumedEnergy_002, ATC_SCE2_PT_MUTotalProducedEnergy_001, ATC_COM_PT_MAPolling_001
[MPC-TS-014]	ATC_COM_PT_MUNotification_001, ATC_COM_PT_MANotification_001

Table 10: Requirement coverage

6.13 Test case coverage

The resulting test case coverage depending on the DUT (test scope) with reference to verdicts is summarized in Table 11. It defines the relevance of conformity of respective DUTs for this document, based upon the type of the DUT (MU or MA) and the instance in which the Use Case is implemented.

The following symbols are used in Table 11:

- "m" represents the mandatory test scope for a specific DUT and instance the test case aims at – the item SHALL be implemented;
- "r" marks recommended test cases – this includes the functional standard which SHOULD be applied; and
- "o" represents the optional test scope – manufacturer MAY decide to implement the item;
- "—" indicates DUTs and instances the test case is not applicable for.

ATC ID	DUT which is covered in the ATC	
	MU	MA
ATC_COM_PT_MUPolling_001	r *[1]	—
ATC_COM_PT_MUNotification_001	r *[1]	—
ATC_SCE1_PT_MUTotalActivePower_001	m	—
ATC_SCE1_PT_MUPhaseActivePower_001	o *[3] *[4]	—
ATC_SCE1_PT_MUPhaseActivePower_002	o *[3] *[4]	—
ATC_SCE1_PT_MUPhaseActivePower_003	o *[3] *[4]	—
ATC_SCE2_PT_MUTotalConsumedEnergy_001	o *[2]	—
ATC_SCE2_PT_MUTotalConsumedEnergy_002	o *[2]	—
ATC_SCE2_PT_MUTotalProducedEnergy_001	o *[2]	—
ATC_SCE2_PT_MUTotalProducedEnergy_002	o *[2]	—
ATC_SCE3_PT_MUActiveACCurrent_001	r *[3] *[4]	—
ATC_SCE3_PT_MUActiveACCurrent_002	r *[3] *[4]	—
ATC_SCE3_PT_MUActiveACCurrent_003	r *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_001	r *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_002	r *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_003	r *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_004	o *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_005	o *[3] *[4]	—
ATC_SCE4_PT_MUACVoltage_006	o *[3] *[4]	—
ATC_SCE5_PT_MUFrequency_001	o *[5]	—
ATC_COM_PT_MAPolling_001	—	r *[1]
ATC_COM_PT_MANotification_001	—	r *[1]
ATC_SCE1_PT_MATotalActivePower_001	—	m
ATC_SCE1_NT_MATotalActivePower_002	—	m

ATC_SCE1_PT_MAPhaseActivePower_001	—	o *[3] *[4]
ATC_SCE1_NT_MAPhaseActivePower_002	—	o *[3] *[4]
ATC_SCE1_PT_MAPhaseActivePower_003	—	o *[3] *[4]
ATC_SCE1_NT_MAPhaseActivePower_004	—	o *[3] *[4]
ATC_SCE1_PT_MAPhaseActivePower_005	—	o *[3] *[4]
ATC_SCE1_NT_MAPhaseActivePower_006	—	o *[3] *[4]
ATC_SCE2_PT_MATotalConsumedEnergy_001	—	o *[2]
ATC_SCE2_NT_MATotalConsumedEnergy_002	—	o *[2]
ATC_SCE2_PT_MATotalProducedEnergy_001	—	o *[2]
ATC_SCE2_NT_MATotalProducedEnergy_002	—	o *[2]
ATC_SCE3_PT_MAActiveACCurrent_001	—	r *[3] *[4]
ATC_SCE3_NT_MAActiveACCurrent_002	—	r *[3] *[4]
ATC_SCE3_PT_MAActiveACCurrent_003	—	r *[3] *[4]
ATC_SCE3_NT_MAActiveACCurrent_004	—	r *[3] *[4]
ATC_SCE3_PT_MAActiveACCurrent_005	—	r *[3] *[4]
ATC_SCE3_NT_MAActiveACCurrent_006	—	r *[3] *[4]
ATC_SCE4_PT_MAACVoltage_001	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_002	—	o *[3] *[4]
ATC_SCE4_PT_MAACVoltage_003	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_004	—	o *[3] *[4]
ATC_SCE4_PT_MAACVoltage_005	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_006	—	o *[3] *[4]
ATC_SCE4_PT_MAACVoltage_007	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_008	—	o *[3] *[4]
ATC_SCE4_PT_MAACVoltage_009	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_010	—	o *[3] *[4]
ATC_SCE4_PT_MAACVoltage_011	—	o *[3] *[4]
ATC_SCE4_NT_MAACVoltage_012	—	o *[3] *[4]
ATC_SCE5_PT_MAFrequency_001	—	o *[5]
ATC_SCE5_NT_MAFrequency_002	—	o *[5]

Table 11: Test case coverage of verdict statements

*[1]: For appliances that request/send data via interval (polling) or notification, the corresponding ATCs are no longer considered "recommended" but "mandatory" and shall be executed.

*[2]: If the data point is supported and the device is capable of either consuming or producing energy the corresponding ATCs are no longer considered "optional" but "mandatory". The manufacturer must specify in the [ParameterSheet] which scenarios are supported.

*[3]: For tests in which the operating phases of the device are tested, the corresponding ATCs are no longer considered "optional" or "recommended" but "mandatory", meaning they verify the phases actually used and shall be executed. The manufacturer must specify in the [ParameterSheet] which measurands are supported.

*[4]: The manufacturer must specify conditions on how the test case can be tested (e.g. via debug interface). For testing, developers must therefore pay attention to these parameters and take them into account in a debug interface.

*[5]: If the data point is supported the corresponding ATCs are no longer considered "optional" or "recommended" but "mandatory". The manufacturer must specify in the [ParameterSheet] which measurands are supported.

7 Abstract test cases for MU

7.1 General information

In this chapter, all ATCs for the Use Case "Monitoring of Power Consumption" conformance test are specified covering the MPC requirements as defined in [MPC1.0.0] and summarized in chapter 5.2 of this document. Within this chapter, the MU is the DUT.

7.2 Common abstract test cases

7.2.1 Messages

Table 12 shows the test case description for ATC_COM_PT_MUPolling_001.

ATC ID	ATC_COM_PT_MUPolling_001
Description	This test shall ensure that the MU sends its data after receiving requests of the MA. The interval between 2 consecutive requests shall not exceed 120 seconds. The tester must choose and document a data point from the manufacturer's specification in the [ParameterSheet].
Referenced Requirement(s)	[MPC-TS-013]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Polling: Request and count 5 data messages sent by the MU.	The MA receives the data of the chosen data point after each request within 120 seconds.

Table 12: ATC_COM_PT_MUPolling_001

Table 13 shows the test case description for ATC_COM_PT_MUNotification_001.

ATC ID	ATC_COM_PT_MUNotification_001
Description	This test shall ensure that the MU sends its data regularly and reliably after a value has changed. After the value change, the changed data shall be transmitted within 120 seconds. *[1] The tester must choose a data point from the manufacturer's specification in the [ParameterSheet].
Referenced Requirement(s)	[MPC-TS-014]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**

1.	Notification: Initiate the value changes on the MU to send data 5 times within 10 minutes.	A new value of the chosen data point is sent by the MU for each initiated value change. Each transmission of the changed data shall be completed within 120 seconds.

Table 13: ATC_COM_PT_MUNotification_001

*[1]: The timeout for the transmission within 120 seconds is to be considered best practice.

7.2.2 Data point total active power

Table 14 shows the test case description for ATC_SCE1_PT_MUTotalActivePower_001.

ATC ID	ATC_SCE1_PT_MUTotalActivePower_001	
Description	This test shall ensure that the MU sends the momentary power consumption/production.	
Referenced Requirement(s)	[MPC-TS-001], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the momentary power consumption/production.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production.	A new value of the momentary power consumption/production is sent by the MU within 120 seconds since the value changed.

Table 14: ATC_SCE1_PT_MUTotalActivePower_001

7.2.3 Data point phase-specific active power

Table 15 shows the test case description for ATC_SCE1_PT_MUPhaseActivePower_001.

ATC ID	ATC_SCE1_PT_MUPhaseActivePower_001
Description	This test shall ensure that the MU sends the phase-specific active power on phase A while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/1], [MPC-TS-002/4], [MPC-TS-009]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce

Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific active power on phase A.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for phase-specific active power on phase A.	A new value for the phase-specific active power on phase A is sent by the MU within 120 seconds since the value changed.

Table 15: ATC_SCE1_PT_MUPhaseActivePower_001

Table 16 shows the test case description for ATC_SCE1_PT_MUPhaseActivePower_002.

ATC ID	ATC_SCE1_PT_MUPhaseActivePower_002	
Description	This test shall ensure that the MU sends the phase-specific active power on phase B while consuming/producing energy.	
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/2], [MPC-TS-002/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific active power on phase B.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for phase-specific active power on phase B.	A new value for the phase-specific active power on phase B is sent by the MU within 120 seconds since the value changed.

Table 16: ATC_SCE1_PT_MUPhaseActivePower_002

Table 17 shows the test case description for ATC_SCE1_PT_MUPhaseActivePower_003.

ATC ID	ATC_SCE1_PT_MUPhaseActivePower_003	
Description	This test shall ensure that the MU sends the phase-specific active power on phase C while consuming/producing energy.	
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/3], [MPC-TS-002/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific active power on phase C.	A value can be determined by the MA.

2.	Initiate a value change on the MU large enough to send data for phase-specific active power on phase C.	A new value for the phase-specific active power on phase C is sent by the MU within 120 seconds since the value changed.

Table 17: ATC_SCE1_PT_MUPhaseActivePower_003

7.2.4 Data point total consumed energy

Table 18 shows the test case description for ATC_SCE2_PT_MUTotalConsumedEnergy_001.

ATC ID	ATC_SCE2_PT_MUTotalConsumedEnergy_001
Description	This test shall ensure that the MU sends a positive total consumed energy while consuming energy.
Referenced Requirement(s)	[MPC-TS-003], [MPC-TS-003/1], [MPC-TS-009]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the total consumed energy.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the total consumed energy.	A new value for the total consumed energy is sent by the MU within 120 seconds since the value changed.

Table 18: ATC_SCE2_PT_MUTotalConsumedEnergy_001

Table 19 shows the test case description for ATC_SCE2_PT_MUTotalConsumedEnergy_002.

ATC ID	ATC_SCE2_PT_MUTotalConsumedEnergy_002
Description	This test shall ensure that the MU sends a positive total consumed energy while producing energy.
Referenced Requirement(s)	[MPC-TS-003], [MPC-TS-003/1], [MPC-TS-009], [MPC-TS-013]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01 Energy directions (all): produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Polling: Request the MU data for the total consumed energy 5 times in a 120 second interval.	A value for the total consumed energy is sent by the MU. The value does not change.

Table 19: ATC_SCE2_PT_MUTotalConsumedEnergy_002

7.2.5 Data point total produced energy

Table 20 shows the test case description for ATC_SCE2_PT_MUTotalProducedEnergy_001.

ATC ID	ATC_SCE2_PT_MUTotalProducedEnergy_001	
Description	This test shall ensure that the MU sends a negative total produced energy while consuming energy.	
Referenced Requirement(s)	[MPC-TS-004], [MPC-TS-004/1], [MPC-TS-009], [MPC-TS-013]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Polling: Request the MU data for the total produced energy 5 times in a 120 second interval.	A value for the total produced energy is sent by the MU. The value does not change.

Table 20: ATC_SCE2_PT_MUTotalProducedEnergy_001

Table 21 shows the test case description for ATC_SCE2_PT_MUTotalProducedEnergy_002.

ATC ID	ATC_SCE2_PT_MUTotalProducedEnergy_002	
Description	This test shall ensure that the MU sends a negative total produced energy while producing energy.	
Referenced Requirement(s)	[MPC-TS-004], [MPC-TS-004/1], [MPC-TS-009]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the total produced energy.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the total produced energy.	A new value for the total produced energy is sent by the MU within 120 seconds since the value changed.

Table 21: ATC_SCE2_PT_MUTotalProducedEnergy_002

7.2.6 Data point phase-specific AC current

Table 22 shows the test case description for ATC_SCE3_PT_MUActiveACCurrent_001.

ATC ID	ATC_SCE3_PT_MUActiveACCurrent_001
Description	This test shall ensure that the MU sends the phase-specific active AC current on phase A while consuming/producing energy.

Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/1], [MPC-TS-005/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific active AC current on phase A.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase A.	A new value for the phase-specific active AC current on phase A is sent by the MU within 120 seconds since the value changed.

648 *Table 22: ATC_SCE3_PT_MUActiveACCurrent_001*

649 Table 23 shows the test case description for ATC_SCE3_PT_MUActiveACCurrent_002.

ATC ID	ATC_SCE3_PT_MUActiveACCurrent_002	
Description	This test shall ensure that the MU sends the phase-specific active AC current on phase B while consuming/producing energy.	
Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/2], [MPC-TS-005/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific active AC current on phase B.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase B.	A new value for the phase-specific active AC current on phase B is sent by the MU within 120 seconds since the value changed.

650 *Table 23: ATC_SCE3_PT_MUActiveACCurrent_002*

651 Table 24 shows the test case description for ATC_SCE3_PT_MUActiveACCurrent_003.

ATC ID	ATC_SCE3_PT_MUActiveACCurrent_003
Description	This test shall ensure that the MU sends the phase-specific active AC current on phase C while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/3], [MPC-TS-005/4], [MPC-TS-009]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested.

	For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific active AC current on phase C.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase C.	A new value for the phase-specific active AC current on phase C is sent by the MU within 120 seconds since the value changed.

Table 24: ATC_SCE3_PT_MUActiveACCurrent_003

7.2.7 Data point phase-specific AC voltage

Table 25 shows the test case description for ATC_SCE4_PT_MUACVoltage_001.

ATC ID	ATC_SCE4_PT_MUACVoltage_001
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase A and neutral.
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/1], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase A and neutral.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase A and neutral.	A new value for the phase-specific AC voltage between phase A and neutral is sent by the MU within 120 seconds since the value changed.

Table 25: ATC_SCE4_PT_MUACVoltage_001

Table 26 shows the test case description for ATC_SCE4_PT_MUACVoltage_002.

ATC ID	ATC_SCE4_PT_MUACVoltage_002
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase B and neutral.
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/2], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

Execution		Expected result
1.	Determine the current MU data for the phase-specific AC voltage between phase B and neutral.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase B and neutral.	A new value for the phase-specific AC voltage between phase B and neutral is sent by the MU within 120 seconds since the value changed.

Table 26: ATC_SCE4_PT_MUACVoltage_002

Table 27 shows the test case description for ATC_SCE4_PT_MUACVoltage_003.

ATC ID	ATC_SCE4_PT_MUACVoltage_003	
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase C and neutral.	
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/3], [MPC-TS-006/7], [MPC-TS-010]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase C and neutral.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase C and neutral.	A new value for the phase-specific AC voltage between phase C and neutral is sent by the MU within 120 seconds since the value changed.

Table 27: ATC_SCE4_PT_MUACVoltage_003

Table 28 shows the test case description for ATC_SCE4_PT_MUACVoltage_004.

ATC ID	ATC_SCE4_PT_MUACVoltage_004	
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase A and phase B.	
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/4], [MPC-TS-006/7], [MPC-TS-010]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase A and phase B.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase A and phase B.	A new value for the phase-specific AC voltage between phase A and phase B is sent by the

		MU within 120 seconds since the value changed.

Table 28: ATC_SCE4_PT_MUACVoltage_004

Table 29 shows the test case description for ATC_SCE4_PT_MUACVoltage_005.

ATC ID	ATC_SCE4_PT_MUACVoltage_005	
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase B and phase C.	
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/5], [MPC-TS-006/7], [MPC-TS-010]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase B and phase C.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase B and phase C.	A new value for the phase-specific AC voltage between phase B and phase C is sent by the MU within 120 seconds since the value changed.

Table 29: ATC_SCE4_PT_MUACVoltage_005

Table 30 shows the test case description for ATC_SCE4_PT_MUACVoltage_006.

ATC ID	ATC_SCE4_PT_MUACVoltage_006	
Description	This test shall ensure that the MU sends the phase-specific AC voltage between phase C and phase A.	
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/6], [MPC-TS-006/7], [MPC-TS-010]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase C and phase A.	A value can be determined by the MA.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase C and phase A.	A new value for the phase-specific AC voltage between phase C and phase A is sent by the MU within 120 seconds since the value changed.

Table 30: ATC_SCE4_PT_MUACVoltage_006

7.2.8 Data point AC frequency

Table 31 shows the test case description for ATC_SCE5_PT_MUFrequency_001.

<table>
<tr><th colspan="2">ATC ID</th><td colspan="2">ATC_SCE5_PT_MUFrequency_001</td></tr>
<tr><th colspan="2">Description</th><td colspan="2">This test shall ensure that the MU sends the frequency.</td></tr>
<tr><th colspan="2">Referenced Requirement(s)</th><td colspan="2">[MPC-TS-007]</td></tr>
<tr><th colspan="2">Test variation</th><td colspan="2">No variation of the setup results into 1 test execution with 1 actor being tested.

For test steps 1 and 2:
Monitoring value states (all): MonitoringState_01</td></tr>
<tr><th colspan="2">Pre-condition</th><td colspan="2">CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished</td></tr>
<tr><th colspan="3">Execution</th><th>Expected result</th></tr>
<tr><td>1.</td><td colspan="2">Determine the current MU data for the frequency.</td><td>A value can be determined by the MA.</td></tr>
<tr><td>2.</td><td colspan="2">Initiate a value change on the MU large enough to send data for the frequency.</td><td>A new value for the frequency is sent by the MU within 120 seconds since the value changed.</td></tr>
</table>

Table 31: ATC_SCE5_PT_MUFrequency_001

8 Abstract test cases for MA

8.1 General information

In this chapter, all ATCs for the Use Case "Monitoring of Power Consumption" conformance test are specified covering the MPC requirements as defined in [MPC1.0.0] and summarized in chapter 5.2 of this document. Within this chapter, the MA is the DUT.

8.2 Common abstract test cases

8.2.1 Messages

Table 32 shows the test case description for ATC_COM_PT_MAPolling_001.

ATC ID	ATC_COM_PT_MAPolling_001	
Description	This test shall ensure that the MA requests data from the MU regularly. The interval between 2 consecutive requests must be specified in the [ParameterSheet].	
Referenced Requirement(s)	[MPC-TS-013]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Polling: Count 5 requests sent by the MA.	The MU receives 5 requests of the MA within the interval specified in the [ParameterSheet].

Table 32: ATC_COM_PT_MAPolling_001

Table 33 shows the test case description for ATC_COM_PT_MANotification_001.

ATC ID	ATC_COM_PT_MANotification_001	
Description	This test shall ensure that the MA receives a notification after a value has changed. After the value change, the changed data shall be transmitted within 120 seconds. *[1]	
Referenced Requirement(s)	[MPC-TS-014]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Notification: Initiate 5 value changes to send data 5 times within 10 minutes.	The MA receives the data. Each transmission of the changed data shall be completed within 120 seconds.

Table 33: ATC_COM_PT_MANotification_001

*[1]: The timeout for the transmission within 120 seconds is to be considered best practice.

8.2.2 Data point total active power

Table 34 shows the test case description for ATC_SCE1_PT_MATotalActivePower_001.

ATC ID	ATC_SCE1_PT_MATotalActivePower_001	
Description	This test shall ensure that the MA receives the momentary power consumption/production with value state "normal".	
Referenced Requirement(s)	[MPC-TS-001], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the momentary power consumption/production.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production.	The MA receives the data for the momentary power consumption/production within 120 seconds since the value changed.

Table 34: ATC_SCE1_PT_MATotalActivePower_001

Table 35 shows the test case description for ATC_SCE1_NT_MATotalActivePower_002.

ATC ID	ATC_SCE1_NT_MATotalActivePower_002	
Description	This test shall ensure that the MA receives the momentary power consumption/production with value state "error" or "out of range".	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Send the MU data for the momentary power consumption/production.	The MA receives the data for the momentary power consumption/production with a value state "error" or "out of range" and discards it.

Table 35: ATC_SCE1_NT_MATotalActivePower_002

8.2.3 Data point phase-specific active power

Table 36 shows the test case description for ATC_SCE1_PT_MAPhaseActivePower_001.

ATC ID	ATC_SCE1_PT_MAPhaseActivePower_001	
Description	This test shall ensure that the MA receives the momentary power consumption/production on phase A with value state "normal".	
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/1], [MPC-TS-002/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the momentary power consumption/production on phase A.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production on phase A.	The MA receives the data for the momentary power consumption/production on phase A within 120 seconds since the value changed.

Table 36: ATC_SCE1_PT_MAPhaseActivePower_001

Table 37 shows the test case description for ATC_SCE1_NT_MAPhaseActivePower_002.

ATC ID	ATC_SCE1_NT_MAPhaseActivePower_002	
Description	This test shall ensure that the MA receives the momentary power consumption/production on phase A with value state "error" or "out of range".	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Send the MU data for the momentary power consumption/production on phase A.	The MA receives the data for the momentary power consumption/production on phase A with a value state "error" or "out of range" and discards it.

Table 37: ATC_SCE1_NT_MAPhaseActivePower_002

Table 38 shows the test case description for ATC_SCE1_PT_MAPhaseActivePower_003.

ATC ID	ATC_SCE1_PT_MAPhaseActivePower_003

Description	This test shall ensure that the MA receives the momentary power consumption/production on phase B with value state "normal".	
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/2], [MPC-TS-002/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the momentary power consumption/production on phase B.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production on phase B.	The MA receives the data for the momentary power consumption/production on phase B within 120 seconds since the value changed.

Table 38: ATC_SCE1_PT_MAPhaseActivePower_003

Table 39 shows the test case description for ATC_SCE1_NT_MAPhaseActivePower_004.

ATC ID	ATC_SCE1_NT_MAPhaseActivePower_004	
Description	This test shall ensure that the MA receives the momentary power consumption/production on phase B with value state "error" or "out of range".	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Send the MU data for the momentary power consumption/production on phase B.	The MA receives the data for the momentary power consumption/production on phase B with a value state "error" or "out of range" and discards it.

Table 39: ATC_SCE1_NT_MAPhaseActivePower_004

Table 40 shows the test case description for ATC_SCE1_PT_MAPhaseActivePower_005.

ATC ID	ATC_SCE1_PT_MAPhaseActivePower_005
Description	This test shall ensure that the MA receives the momentary power consumption/production on phase C with value state "normal".
Referenced Requirement(s)	[MPC-TS-002], [MPC-TS-002/3], [MPC-TS-002/4], [MPC-TS-009]

Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the momentary power consumption/production on phase C.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the momentary power consumption/production on phase C.	The MA receives the data for the momentary power consumption/production on phase C within 120 seconds since the value changed.

Table 40: ATC_SCE1_PT_MAPhaseActivePower_005

Table 41 shows the test case description for ATC_SCE1_NT_MAPhaseActivePower_006.

ATC ID	ATC_SCE1_NT_MAPhaseActivePower_006	
Description	This test shall ensure that the MA receives the momentary power consumption/production on phase C with value state "error" or "out of range".	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Send the MU data for the momentary power consumption/production on phase C.	The MA receives the data for the momentary power consumption/production on phase C with a value state "error" or "out of range" and discards it.

Table 41: ATC_SCE1_NT_MAPhaseActivePower_006

8.2.4 Data point total consumed energy

Table 42 shows the test case description for ATC_SCE2_PT_MATotalConsumedEnergy_001.

ATC ID	ATC_SCE2_PT_MATotalConsumedEnergy_001
Description	This test shall ensure that the MA receives the total consumed energy with value state "normal" while consuming energy.
Referenced Requirement(s)	[MPC-TS-003], [MPC-TS-003/1], [MPC-TS-009]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested.

	For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Determine the current MU data for the total consumed energy.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the total consumed energy.	The MA receives the data for the total consumed energy within 120 seconds since the value changed.

Table 42: ATC_SCE2_PT_MATotalConsumedEnergy_001

Table 43 shows the test case description for ATC_SCE2_NT_MATotalConsumedEnergy_002.

ATC ID	ATC_SCE2_NT_MATotalConsumedEnergy_002	
Description	This test shall ensure that the MA receives the total consumed energy with value state "error" or "out of range" while consuming energy.	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**
1.	Send the MU data for the total consumed energy.	The MA receives the data for the total consumed energy with a value state "error" or "out of range" and discards it.

Table 43: ATC_SCE2_NT_MATotalConsumedEnergy_002

8.2.5 Data point total produced energy

Table 44 shows the test case description for ATC_SCE2_PT_MATotalProducedEnergy_001.

ATC ID	ATC_SCE2_PT_MATotalProducedEnergy_001	
Description	This test shall ensure that the MA receives the total produced energy with value state "normal" while producing energy.	
Referenced Requirement(s)	[MPC-TS-004], [MPC-TS-004/1], [MPC-TS-009]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
	Execution	**Expected result**

1.	Determine the current MU data for the total produced energy.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the total produced energy.	The MA receives the data for the total produced energy within 120 seconds since the value changed.

Table 44: ATC_SCE2_PT_MATotalProducedEnergy_001

Table 45 shows the test case description for ATC_SCE2_NT_MATotalProducedEnergy_002.

ATC ID	ATC_SCE2_NT_MATotalProducedEnergy_002	
Description	This test shall ensure that the MA receives the total produced energy with value state "error" or "out of range" while producing energy.	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Send the MU data for the total produced energy.	The MA receives the data for the total produced energy with a value state "error" or "out of range" and discards it.

Table 45: ATC_SCE2_NT_MATotalProducedEnergy_002

8.2.6 Data point phase-specific AC current

Table 46 shows the test case description for ATC_SCE3_PT_MAActiveACCurrent_001.

ATC ID	ATC_SCE3_PT_MAActiveACCurrent_001	
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase A with value state "normal" while consuming/producing energy.	
Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/1], [MPC-TS-005/4], [MPC-TS-009]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific active AC current on phase A.	The MA already has or receives the determined value.

2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase A.	The MA receives the data for the phase-specific active AC current on phase A within 120 seconds since the value changed.

Table 46: ATC_SCE3_PT_MAActiveACCurrent_001

Table 47 shows the test case description for ATC_SCE3_NT_MAActiveACCurrent_002.

ATC ID	ATC_SCE3_NT_MAActiveACCurrent_002
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase A with value state "error" or "out of range" while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific active AC current on phase A.	The MA receives the data for the phase-specific active AC current on phase A with a value state "error" or "out of range" and discards it.

Table 47: ATC_SCE3_NT_MAActiveACCurrent_002

Table 48 shows the test case description for ATC_SCE3_PT_MAActiveACCurrent_003.

ATC ID	ATC_SCE3_PT_MAActiveACCurrent_003
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase B with value state "normal" while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/2], [MPC-TS-005/4], [MPC-TS-009]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific active AC current on phase B.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase B.	The MA receives the data for the phase-specific active AC current on phase B within 120 seconds since the value changed.

Table 48: ATC_SCE3_PT_MAActiveACCurrent_003

Table 49 shows the test case description for ATC_SCE3_NT_MAActiveACCurrent_004.

ATC ID	ATC_SCE3_NT_MAActiveACCurrent_004
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase B with value state "error" or "out of range" while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	Expected result
1.	Send the MU data for the phase-specific active AC current on phase B.	The MA receives the data for the phase-specific active AC current on phase B with a value state "error" or "out of range" and discards it.

726 *Table 49: ATC_SCE3_NT_MAActiveACCurrent_004*

727 Table 50 shows the test case description for ATC_SCE3_PT_MAActiveACCurrent_005.

ATC ID	ATC_SCE3_PT_MAActiveACCurrent_005
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase C with value state "normal" while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-005], [MPC-TS-005/3], [MPC-TS-005/4], [MPC-TS-009]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	Expected result
1.	Determine the current MU data for the phase-specific active AC current on phase C.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific active AC current on phase C.	The MA receives the data for the phase-specific active AC current on phase C within 120 seconds since the value changed.

728 *Table 50: ATC_SCE3_PT_MAActiveACCurrent_005*

729 Table 51 shows the test case description for ATC_SCE3_NT_MAActiveACCurrent_006.

ATC ID	ATC_SCE3_NT_MAActiveACCurrent_006
Description	This test shall ensure that the MA receives the phase-specific active AC current on phase C with value state "error" or "out of range" while consuming/producing energy.
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]

Test variation	The variation of the data sets result into a sum of 4 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03 Energy directions (all): consume, produce
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific active AC current on phase C.	The MA receives the data for the phase-specific active AC current on phase C with a value state "error" or "out of range" and discards it.

Table 51: ATC_SCE3_NT_MAActiveACCurrent_006

8.2.7 Data point phase-specific AC voltage

Table 52 shows the test case description for ATC_SCE4_PT_MAACVoltage_001.

ATC ID	ATC_SCE4_PT_MAACVoltage_001
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase A and neutral with value state "normal".
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/1], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase A and neutral.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase A and neutral.	The MA receives the data for the phase-specific AC voltage between phase A and neutral within 120 seconds since the value changed.

Table 52: ATC_SCE4_PT_MAACVoltage_001

Table 53 shows the test case description for ATC_SCE4_NT_MAACVoltage_002.

ATC ID	ATC_SCE4_NT_MAACVoltage_002
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase A and neutral with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03

Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific AC voltage between phase A and neutral.	The MA receives the data for the phase-specific AC voltage between phase A and neutral with a value state "error" or "out of range" and discards it.

736 *Table 53: ATC_SCE4_NT_MAACVoltage_002*

737 Table 54 shows the test case description for ATC_SCE4_PT_MAACVoltage_003.

ATC ID	ATC_SCE4_PT_MAACVoltage_003
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase B and neutral with value state "normal".
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/2], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase B and neutral.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase B and neutral.	The MA receives the data for the phase-specific AC voltage between phase B and neutral within 120 seconds since the value changed.

738 *Table 54: ATC_SCE4_PT_MAACVoltage_003*

739 Table 55 shows the test case description for ATC_SCE4_NT_MAACVoltage_004.

ATC ID	ATC_SCE4_NT_MAACVoltage_004
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase B and neutral with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific AC voltage between phase B and neutral.	The MA receives the data for the phase-specific AC voltage between phase B and neutral with a value state "error" or "out of range" and discards it.

740 *Table 55: ATC_SCE4_NT_MAACVoltage_004*

741 Table 56 shows the test case description for ATC_SCE4_PT_MAACVoltage_005.

ATC ID	ATC_SCE4_PT_MAACVoltage_005
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase C and neutral with value state "normal".
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/3], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase C and neutral.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase C and neutral.	The MA receives the data for the phase-specific AC voltage between phase C and neutral within 120 seconds since the value changed.

742 *Table 56: ATC_SCE4_PT_MAACVoltage_005*

743 Table 57 shows the test case description for ATC_SCE4_NT_MAACVoltage_006.

ATC ID	ATC_SCE4_NT_MAACVoltage_006
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase C and neutral with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific AC voltage between phase C and neutral.	The MA receives the data for the phase-specific AC voltage between phase C and neutral with a value state "error" or "out of range" and discards it.

744 *Table 57: ATC_SCE4_NT_MAACVoltage_006*

745 Table 58 shows the test case description for ATC_SCE4_PT_MAACVoltage_007.

ATC ID	ATC_SCE4_PT_MAACVoltage_007
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase A and phase B with value state "normal".
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/4], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested.

	For test steps 1 and 2: Monitoring value states (all): MonitoringState_01		
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished		
Execution		**Expected result**	
1.	Determine the current MU data for the phase-specific AC voltage between phase A and phase B.	The MA already has or receives the determined value.	
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase A and phase B.	The MA receives the data for the phase-specific AC voltage between phase A and phase B within 120 seconds since the value changed.	

746 *Table 58: ATC_SCE4_PT_MAACVoltage_007*

747 Table 59 shows the test case description for ATC_SCE4_NT_MAACVoltage_008.

ATC ID	ATC_SCE4_NT_MAACVoltage_008	
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase A and phase B with value state "error" or "out of range".	
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]	
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Send the MU data for the phase-specific AC voltage between phase A and phase B.	The MA receives the data for the phase-specific AC voltage between phase A and phase B with a value state "error" or "out of range" and discards it.

748 *Table 59: ATC_SCE4_NT_MAACVoltage_008*

749 Table 60 shows the test case description for ATC_SCE4_PT_MAACVoltage_009.

ATC ID	ATC_SCE4_PT_MAACVoltage_009	
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase B and phase C with value state "normal".	
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/5], [MPC-TS-006/7], [MPC-TS-010]	
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase B and phase C.	The MA already has or receives the determined value.

2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase B and phase C.	The MA receives the data for the phase-specific AC voltage between phase B and phase C within 120 seconds since the value changed.

750 *Table 60: ATC_SCE4_PT_MAACVoltage_009*

751 Table 61 shows the test case description for ATC_SCE4_NT_MAACVoltage_010.

ATC ID	ATC_SCE4_NT_MAACVoltage_010
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase B and phase C with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Send the MU data for the phase-specific AC voltage between phase B and phase C.	The MA receives the data for the phase-specific AC voltage between phase B and phase C with a value state "error" or "out of range" and discards it.

752 *Table 61: ATC_SCE4_NT_MAACVoltage_010*

753 Table 62 shows the test case description for ATC_SCE4_PT_MAACVoltage_011.

ATC ID	ATC_SCE4_PT_MAACVoltage_011
Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase C and phase A with value state "normal".
Referenced Requirement(s)	[MPC-TS-006], [MPC-TS-006/6], [MPC-TS-006/7], [MPC-TS-010]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	**Expected result**
1.	Determine the current MU data for the phase-specific AC voltage between phase C and phase A.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the phase-specific AC voltage between phase C and phase A.	The MA receives the data for the phase-specific AC voltage between phase C and phase A within 120 seconds since the value changed.

754 *Table 62: ATC_SCE4_PT_MAACVoltage_011*

755 Table 63 shows the test case description for ATC_SCE4_NT_MAACVoltage_012.

ATC ID	ATC_SCE4_NT_MAACVoltage_012

Description	This test shall ensure that the MA receives the phase-specific AC voltage between phase C and phase A with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested. For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	Expected result
1.	Send the MU data for the phase-specific AC voltage between phase C and phase A.	The MA receives the data for the phase-specific AC voltage between phase C and phase A with a value state "error" or "out of range" and discards it.

Table 63: ATC_SCE4_NT_MAACVoltage_012

8.2.8 Data point AC frequency

Table 64 shows the test case description for ATC_SCE5_PT_MAFrequency_001.

ATC ID	ATC_SCE5_PT_MAFrequency_001
Description	This test shall ensure that the MA receives the frequency with value state "normal".
Referenced Requirement(s)	[MPC-TS-007]
Test variation	No variation of the setup results into 1 test execution with 1 actor being tested. For test steps 1 and 2: Monitoring value states (all): MonitoringState_01
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished

	Execution	Expected result
1.	Determine the current MU data for the frequency.	The MA already has or receives the determined value.
2.	Initiate a value change on the MU large enough to send data for the frequency.	The MA receives data for the frequency within 120 seconds since the value changed.

Table 64: ATC_SCE5_PT_MAFrequency_001

Table 65 shows the test case description for ATC_SCE5_NT_MAFrequency_002.

ATC ID	ATC_SCE5_NT_MAFrequency_002
Description	This test shall ensure that the MA receives the frequency with value state "error" or "out of range".
Referenced Requirement(s)	[MPC-TS-008], [MPC-TS-008/1], [MPC-TS-008/2]
Test variation	The variation of the data sets result into a sum of 2 test executions with 1 actor being tested.

	For test step 1: Monitoring value states (all): MonitoringState_02, MonitoringState_03	
Pre-condition	CF_MU_ConnectionEstablished, CF_MA_ConnectionEstablished	
Execution		**Expected result**
1.	Send the MU data for the frequency.	The MA receives the data for the frequency with a value state "error" or "out of range" and discards it.

Table 65: ATC_SCE5_NT_MAFrequency_002